

Worksheet 9: probabilistic forecasting

Slide 5

$$\text{Brier score} = \frac{1}{N} \sum_i (p_i - y_i)^2$$

$$y_i = \begin{cases} 0 & \text{no rain} \\ 1 & \text{rain} \end{cases}$$

Days	Forecast	Obs. y_i	$(p_i - y_i)^2$
1	0.2	0	0.04
2	0.7	1	0.09
3	0.1	0	0.01
4	0.9	1	0.01
5	0.4	0	0.16

$$BS = \frac{0.31}{5} = 0.06$$

You could compare this with another forecast model to see which is better.

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$$\begin{aligned} LS(\mu, \sigma) &= -\log f(y) \\ &= +\frac{1}{2}\log(2\pi\sigma^2) + \frac{(y - \mu)^2}{2\sigma^2} \end{aligned}$$

Suppose observation $y = 12$ and forecast $\mu = 10, \sigma = 3$

$$\begin{aligned} LS &= \frac{1}{2}\log(2\pi 3^2) + \frac{(12 - 10)^2}{2 \times 3^2} \\ &= 2.24 \end{aligned}$$

Again $\rightarrow$ compare with another forecast.

We would average these over multiple forecasts.

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$$CRPS(\mu, \sigma, y) = \sigma \left[z(2\Phi(z) - 1) + 2\phi(z) - \frac{1}{\sqrt{\pi}} \right]$$

where Φ is the normal cdf and ϕ is the pdf.

$$z = \frac{x - \mu}{\sigma}$$

Observation $y = 12, \mu = 10, \sigma = 3$

$$z = \frac{12 - 10}{3} = \frac{2}{3}$$

$$\Phi(z) = 0.7475, \quad \phi(z) = 0.3194$$

$$\begin{aligned} CRPS &= 3 \left[\frac{2}{3} (2 \times 0.7475 - 1) + 2 \times 0.3194 - \frac{1}{\sqrt{\pi}} \right] \\ &= 1.214 \end{aligned}$$

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from scipy.stats import norm
norm.cdf
norm.pdf
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Slide 9

$$\begin{bmatrix} \mu \\ \sigma \end{bmatrix} = \begin{bmatrix} 0.32 \\ -0.242 \end{bmatrix}$$

so $\mu = 0.32$

$$\begin{aligned} \sigma &= \text{softplus}(-0.242) \\ &= \log(1 + e^{-0.242}) \\ &= 0.251 \end{aligned}$$

$$\begin{aligned} \text{Note } \sigma &= \log(1 + e^s) \\ e^\sigma &= e^s - 1 \\ s &= \log(e^\sigma - 1) \end{aligned}$$

Suppose $x = \begin{bmatrix} 0.5 & -0.2 \end{bmatrix}$

$$W = \begin{bmatrix} 0.8 & -0.4 \\ 0.3 & 0.6 \end{bmatrix}, \quad b = \begin{bmatrix} 0.1 \\ -0.05 \end{bmatrix}$$

$$z = Wx + b = \begin{bmatrix} 0.8 & -0.4 \\ 0.3 & 0.6 \end{bmatrix} \begin{bmatrix} 0.5 \\ -0.2 \end{bmatrix} + \begin{bmatrix} 0.1 \\ -0.05 \end{bmatrix} = \begin{bmatrix} 0.58 \\ -0.02 \end{bmatrix}$$

$$h = \text{ReLU} \begin{bmatrix} 0.58 \\ -0.02 \end{bmatrix} = \begin{bmatrix} 0.58 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \mu \\ \sigma \end{bmatrix} = W_y h + b = \begin{bmatrix} 0.5 & -0.2 \\ 0.1 & 0.3 \end{bmatrix} \begin{bmatrix} 0.58 \\ 0 \end{bmatrix} + \begin{bmatrix} 0.03 \\ -0.3 \end{bmatrix} = \begin{bmatrix} 0.32 \\ -0.242 \end{bmatrix}$$